

Jacob TerMaat

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EDUCATION

Purdue University, West Lafayette, IN

2022 – Present

Bachelor of Science — Computer Engineering; Mathematics Minor

Concentration in Computer Systems

GPA: 3.61 — Dean's List - Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024 & Fall 2025

Graduation: Anticipated May 2026

ACADEMIC PROJECTS

FPGA-Tested Dual Core Processor

August 2025 – December 2025

- Architected and implemented a custom RISC-V processor, developing the ALU and register files while engineering a high-performance control unit to manage pipeline stages and memory-mapped I/O for FPGA execution.
- Improved clock frequency and throughput 300% through pipelining, cache hierarchy, and MSI dual-core coherence.
- Validated architectural correctness and timing using SystemVerilog testbenches, developing comprehensive functional coverage to identify and resolve execution edge cases, and synchronization hazards.

AI-Driven, Computer Vision, Euchre Playing Robot

August 2025 – Present

- Developing an interdisciplinary senior design project focused on building an AI-driven Euchre-playing robot.
- Leading integration of computer vision for card recognition, robotic actuation for gameplay, and rule-based decision-making with planned reinforcement learning enhancements.
- Coordinated hardware, software, and computer vision components into a real-time embedded system.

RELEVANT WORK AND LEADERSHIP EXPERIENCE

IMMI (Indiana Mills & Manufacturing Incorporated), Westfield, IN - R&D Intern

May – August 2025

- Engineered and prototyped a seatbelt system with detection capabilities to enhance vehicle safety.
 - Created a patent-pending design occupant detection design using a retractor, adaptable to various industries.
- Developed a sensor-enabled buckle for real-time seatbelt engagement detection and accurate data.
- Designed standalone devices to rapidly acquire, log, and analyze acceleration data for occupant safety.
- Collaborated with R&D teams to accelerate development of next-generation occupant safety systems.

WeatherTech, Bolingbrook, IL - Tooling Department Intern

May – Aug 2023,24

- Assisted in the tooling process to ensure high quality aluminum molds that were laser precise.
- Maintained and operated large-scale CNC milling machines, along with aligning and balancing tools.
- Learned machine programming for the specialized CNC operations and the arts of tool making.

EPICS, West Lafayette, IN - Project Manager

Aug 2022 – May 2025

- Partnered with local community organizations to identify needs and deliver solutions for the Purdue community.
- Lead teams to deliver community-focused solutions on schedule by managing workflows and project deadlines.

TECHNICAL SKILLS

Computer Programming Languages - Proficient in C, Python, Verilog, SystemVerilog & MATLAB, Experience in Java

CAD Softwares - Proficient in Solidworks, NX, AutoCAD, SketchUp & TinkerCAD

RELEVANT COURSEWORK

Computer Design and Prototyping, ASIC Design Laboratory, Microprocessor Systems and Interfacing, Data Structures & Algorithms, Computer Security, Computer Graphics, Probabilistic Methods, Signals and Systems, Digital System Design, Real Analysis, Discrete Mathematics